

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

74.8 - 2 marks from B₁ and
1, 2, 3 - 1 mark, always mark.

MARKING SCHEME

NOVEMBER 2018

PURE MATHEMATICS

6042/1

B1

1 One root $x = -2i$

$$f(2) = 8 - 8 + 8 - 8 = 0$$

$$(x - 2)(x^2 + 4) = 0 \text{ or equivalent}$$

M1

One factor is $x - 2 \Rightarrow x = 2$

A1

[3]

2 $\cos(\theta - 75^\circ) = \frac{\sqrt{6} - \sqrt{2}}{4}$

$$\therefore \theta - 75^\circ = \pm 75^\circ$$

$$\left(\cos^{-1}\left(\frac{16-19}{4}\right)\right)$$

M1

$$\theta = 0^\circ; \text{ or } 150^\circ; \text{ or } 360^\circ$$

For any one correct

A1

For all correct

A1

[3]

3 $e^{2x+y} = 1$

$$\therefore 2x + y = 0 \dots (1)$$

B1

$$4x - 3y = 10 \dots (2)$$

$$\therefore 10x = 10 \text{ or equivalent solving simultaneously}$$

M1

$$x = 1 \text{ and } y = -2 \text{ for both correct}$$

A1

[3]

4 $\frac{21+13x}{2+3x^2} \geq 10$

$$\therefore 21 + 13x \geq 20 + 30x^2$$

attempt

single or cross multiply

M1

$$0 \geq 30x^2 - 13x - 1 \text{ for quadratic}$$

correct quadratic

A1

$$(15x + 1)(2x - 1) \leq 0 \text{ or a.e.}$$

solving quadratic

M1

$$\therefore -\frac{1}{15} \leq x \leq \frac{1}{2} \text{ for c.a.}$$

A1

[4]

5 $y^2 = 3xy + x^2 - 3$

$$2y \frac{dy}{dx} = 3x \frac{dy}{dx} + 3y + 2x$$

all correct

M1 A1

$$\frac{dy}{dx} = \frac{3y+2x}{2y-3x}$$

when $y = 1$, $1 = 3x + x^2 - 3$ *subst $y=1$*

$$x^2 + 3x - 4 = 0$$

$$(x+4)(x-1) = 0$$

$$x = -4 \text{ or } 1 \text{ for } x = 1$$

at $(1,1)$ $\frac{dy}{dx} = \frac{3+2}{2-3} = -5$

$\therefore$ equation is $\frac{y-1}{x-1} = -5$ *finding eqn*

$$y = -5x + 6$$

or a.e.

M1

M1

A1

B1

M1

A1 [8]

6 (i) $f(x) = -x^2 + 8x - 12 = -(x^2 - 8x + 12)$

$$= -[(x-4)^2 - 16 + 12] \text{ for completing square or a.e.} \rightarrow \text{M1}$$

$$= -(x-4)^2 + 4 \text{ for all a, b, c correct} \rightarrow \text{A1 (2)}$$

(ii) The function is one-to-one or an equivalent explanation *horizontal line test*

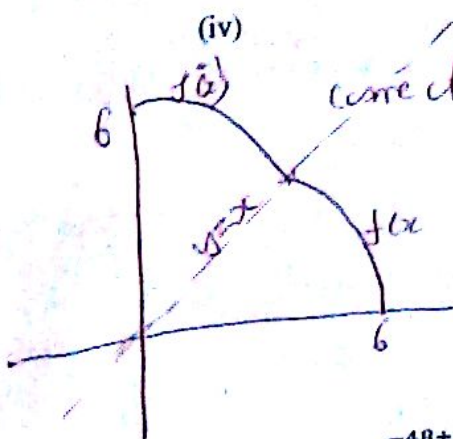
B1

(iii) $y = -(x-4)^2 + 4$

$$(x-4)^2 = 4-y \text{ try to make } x \text{ subj} \rightarrow \text{M1}$$

$$f^{-1}(x) = 4 + \sqrt{4-x} \text{ with or without domain correct ans} \rightarrow \text{A1 (3)}$$

$$\text{Domain of } f(x)^{-1} \text{ is } 0 \leq x \leq 4 \text{ B1}$$



correct f(x) graph including correct intercepts

B1

correct inverse f(x) with correct intercept

B1

[8]

7 (a)

$$\frac{-48+56t}{u} = 1 + 13t$$

$$u = \frac{-48+56t}{1+13t} \text{ attempt to make } u \text{ subject or a.e.}$$

B7

$$= \frac{(-48+56i)(1-13i)}{(1+13i)(1-13i)} \text{ for realising denominator} \quad \text{M1}$$

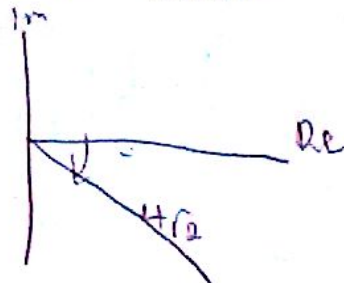
$$= \frac{-48+624i+56i+728}{1+169} \text{ use of } i^2 \text{ for correct unsimplified} \quad \text{M1}$$

$$4+4i$$

A1 (4)

(b) (i) $u^* = 4-4i$ for his u^*

B1✓



(ii)

Correct line in quad ^{line correct} B1✓

$$|u^*| = 4\sqrt{2} \quad \text{B1✓}$$

$$\arg u^* = -\frac{\pi}{4} \text{ Rad.} \quad \text{B1✓} \quad [8]$$

8 (a) $\int x e^{-2x} dx = \frac{-1}{2} x e^{-2x} + \int \frac{1}{2} e^{-2x} dx$ by parts correctly

M1 A1

$$= \frac{-1}{2} x e^{-2x} - \frac{1}{4} e^{-2x} + c$$

A1 (2)

(b) $V = \pi \int_0^1 (2x + e^{-2x})^2 dx$ correct structure

B1

$$= \pi \int_0^1 (4x^2 + 4x e^{-2x} + e^{-4x}) dx$$

$$= \pi \left[\frac{4}{3} x^3 + 4 \left(-\frac{1}{2} x e^{-2x} - \frac{1}{4} e^{-2x} \right) - \frac{1}{4} e^{-4x} \right]_0^1 \text{ expanding and}$$

attempt integrating M1
For integration correctly M1A1

$$= \pi \left\{ \frac{4}{3} - 2e^{-2} - e^{-2} - \frac{1}{4} e^{-4} - \left(-1 - \frac{1}{4} \right) \right\} \text{ Use of limits} \quad \text{M1}$$

9 (a) $\frac{15x^2+40x+2}{(4+x)(2+5x^2)} = \frac{A}{4+x} + \frac{Bx+c}{2+5x^2}$ $= 6.83$ 3 s.f. $[9]$
structure B1

$$15x^2 + 40x + 2 = A(2 + 5x^2) + (Bx + c)(4 + x)$$

$$x = -4; 240 - 160 + 2 = 82A$$

$$82 = 82A \Rightarrow A = 1$$

method of find 1st constant.

M1

A1

$$x = 0 \Rightarrow 2 = 2A + 4c \Rightarrow c = 0 \quad \text{A1}$$

$$x = 1 \Rightarrow 15 + 40 + 2 = 7A(B+c) \Rightarrow B = 10 \quad \text{A1}$$

$$\frac{1}{4+x} + \frac{10x}{2+5x^2} \quad \text{9)$$

$$(b) \int_0^2 \left(\frac{1}{4+x} + \frac{10x}{2+5x^2} \right) dx$$

$$[\ln(4+x) + \ln(2+5x^2))]_0^2$$

attempt to integ. correct integration $\rightarrow$ M1 5-1

$$\ln(6(22)) - \ln 4 \times 2$$

Use of laws of logs $\rightarrow$ M1

$$= \ln\left(\frac{6 \times 22}{4}\right) = \ln\left(\frac{33}{2}\right) \quad \text{exact} \quad \text{A1}^{(4)} [9]$$

$$10 \quad (a) \quad (i) \quad \vec{OA} = 4\lambda i - 3\lambda + 4\lambda k \quad \vec{OB} = 3i + 9j - 6k$$

$$\vec{AB} = (3+4\lambda)i + (9+3\lambda)j + (-6-4\lambda)k \quad \text{or } \vec{BA} \quad \text{B1}$$

$$\vec{AB} \cdot \vec{OB} = 0$$

$$3(3+4\lambda) + 9(9+3\lambda) + 6(6+4\lambda) = 0 \quad \text{use of dot product} \quad \text{M1}$$

$$63\lambda = -126 \Rightarrow \lambda = -2 \quad \rightarrow \text{A1} \quad (3)$$

$$(ii) \quad \vec{AO} = -8i - 6j + 8k; \quad \vec{BO} = -3i - 9j + 6k \quad \text{or equiv.} \quad \text{B1 for both}$$

$$\cos AOB = \frac{24+54+48}{\sqrt{64+36+64} \sqrt{9+81+36}} \quad \text{all correct} \quad \text{M1}$$

$$= \frac{126}{\sqrt{20664}} \Rightarrow AOB = 28.8^\circ \quad \text{or } 29 \quad \text{A1}^{(2)}$$

$$(b) \quad (i) \quad \vec{OA} = 8i + 6j - 8k$$

Mid OA is (4, 3, -4) for centre

double radius correct centre $\rightarrow$ M1 A1

$$(ii) \quad |\vec{OA}| = \sqrt{16+9+16}$$

$$\text{Radius} = \sqrt{41} \quad \text{or } 6.40$$

A1^{(4)} [9]

11 (a) $(1+ax)^n = 1 + nax + \frac{n(n-1)}{2}a^2x^2 + \frac{n(n-1)(n-2)}{6}a^3x^3$ Use of binom expansion at least 3 terms
Explanation M1 M1
all correct

$$na = -1 \dots (1)$$

$$\frac{n(n-1)(n-2)}{6}a^3 = -\frac{14}{3} \quad (2) \text{ for both}$$

one correct eqn. $\rightarrow$ A1

$$(1) \frac{n(n-1)(n-2)}{6} \left(\frac{-1}{n}\right)^3 = -\frac{14}{3}$$

solving eqns. $\rightarrow$ M1

$$(n-1)(n-2) = 6n^2 \times \frac{14}{3}$$

$$n^2 - 3n + 2 = 28n^2$$

obtain eqn genuinely $\rightarrow$ A1
A.G.

$$0 = 27n^2 + 3n - 2$$

[4]

(b) (i) $27n^2 + 3n - 2 = 0$

$$(3n+1)(9n-2) = 0$$

solving qudd. $\rightarrow$ M1
For attempt to solve

$$n = -\frac{1}{3}, n = \frac{2}{9}$$

only. $\rightarrow$ A1

(ii) $-\frac{1}{3}a = -1$

subst value of n. $\rightarrow$ M1

$$\Rightarrow a = 3$$

$\rightarrow$ A1

(iii) Valid for $|3x| < 1$

$$|x| < \frac{1}{3}$$

B1 (5) [9]

12 (a) $f(x) = x^3 - 4e^{\frac{1}{2}x}$

$$f(1.9) = -3.48383864$$

subst. $\rightarrow$ M1

$$f(2.5) = 1.66362817$$

for both correct at least 1 correct $\rightarrow$ A1

There is change of sign, hence a root between them. Correct conclusion A1 (3)

(b) $f'(x) = 3x^2 - 2e^{\frac{1}{2}x}$

$\rightarrow$ B1

$$x_{n+1} = x_n - \frac{x_n^3 - 4e^{\frac{1}{2}x_n}}{3x_n^2 - 2e^{\frac{1}{2}x_n}}$$

structure $\rightarrow$ M1

$$= \frac{3x_n^3 - 2x_n e^{\frac{1}{2}x_n} - x_n^{\frac{1}{2}} + 4e^{\frac{1}{2}x_n}}{3x_n^3 - 2e^{\frac{1}{2}x_n}}$$

$$= \frac{2x_n^3 + 4e^{\frac{1}{2}x_n} - 2x_n e^{\frac{1}{2}x_n}}{3x_n^3 - 2e^{\frac{1}{2}x_n}} \quad \text{single denominator and grouping} \quad \text{M1}$$

$$= \frac{2x_n^3 + 2e^{\frac{1}{2}x_n}(2 - x_n)}{3x_n^3 - 2e^{\frac{1}{2}x_n}} \quad \text{AQ} \quad \text{A1 (2)}$$

(c) $x_1 = 2$

$$\therefore x_2 = \frac{2(2)^3 + 2e(2-2)}{3(2)^3 - 2e} \quad \text{substituting} \quad \text{M1}$$

$$= 2.438 \quad \text{3 d.p.} \quad \text{A1 (2) [9]}$$

13 (a) (i) $a + 4d + a + 9d = 27$

$$2a + 13d = 27 \dots (1)$$

$$a + 2d = -9 \dots (2) \quad \text{For any correct equation} \quad \text{one}$$

$$2a + 4d = -18 \dots (3) \quad \text{B1}$$

$$9d = 45 \quad \text{for solving simultaneously} \quad \text{M1}$$

$$d = 5 \quad \text{A1}$$

$$\text{Subst. in (2), } a + 10 = -9$$

$$a = -19 \quad \text{A1 (4)}$$

(ii) $\frac{n}{2} \{-38 + 5n - 5\} > 0 \quad \text{substituting and cancelling} \quad \text{M1 A1}$

$$n(-43 + 5n) > 0 \quad \text{A1}$$

$$n = 9 \quad \text{For critical values} \quad \text{A1 (3)}$$

(b) (i) $\frac{ar^5}{ar^3} = \frac{1.6348}{2.56} \Rightarrow r^2 = 0.63859375 \quad \text{M1}$

$$\therefore r = -0.79912061 \quad \text{and} \quad \text{A1}$$

$$a = -5 \quad -5.026525067 \quad \text{A1 (3)}$$

$$(ii) S_{\infty} = \frac{-5}{1+0.8} \quad \text{M1}$$

$$= -2\frac{7}{9} \text{ or } -2.77 \quad \text{A1 (2) [12]}$$

- 14 (a) $A(-3, k); B(3, 3k);$ gradient of radius through B is $\frac{4}{3}$

$$\therefore \frac{3k-k}{3+3} = \frac{4}{3} \quad \text{finding } k \quad \text{M1}$$

$$\frac{2k}{6} = \frac{4}{3} \Rightarrow k = 4 \quad \text{A1}$$

- (b) (i) $A(-3, 4); B(3, 12);$

$$r^2 = (3+3)^2 + (12-4)^2 = 100 \text{ or a.e.} \quad \text{B1}$$

$$(x+3)^2 + (y-4)^2 = 100 \quad \text{finding eqn of circle} \quad \text{M1}$$

$$x^2 + 6x + 9 + y^2 - 8y + 16 = 100$$

$$x^2 + y^2 + 6x - 8y - 75 = 0 \quad \text{A1}$$

$$(ii) y = 0 \text{ on } x\text{-axis} \Rightarrow x^2 + 6x - 75 = 0 \quad \text{B1} \checkmark$$

$$x = \frac{-6 \pm \sqrt{36+300}}{2} \quad \text{solving} \quad \text{M1}$$

$$= -12.17 \text{ or } 6.17 \text{ for both} \quad \text{A1}$$

- (iii) $C(-12, 17, 0);$ and $D(6, 17, 0);$

$$\therefore CD = 6.17 + 12.17 = 18.34 \quad \text{B1} \checkmark$$

$$\cos \widehat{CAD} = \frac{10^2 + 10^2 - (18.34)^2}{200} \quad \text{finding angle} \quad \text{M1}$$

$$\widehat{CAD} = 2.32 \text{ radians or } 132.9^{\circ} \text{ s.o.i} \quad \text{A1}$$

$$\text{Area} = \frac{1}{2} \times 10^2 \times 2.32 - \frac{1}{2} \times 10^2 \sin 2.32 \quad \text{finding area} \quad \text{M1}$$

$$= 79.39 \quad \text{A1 [13]}$$

- 15 (a) $k = 3$

B1

(b) $y = \frac{1}{32}x + \frac{1}{2}(x-3)^{-1}$

$$\frac{dy}{dx} = \frac{1}{32} - \frac{1}{2}(x-3)^{-2}$$

equating to zero
M1 A1
correct derivative

$$\text{At turning points} = \frac{1}{32} - \frac{1}{2} \cdot \frac{1}{(x-3)^2} = 0$$

$$(x-3)^2 = 16 \quad \text{equating to 0 and attempting to solve}$$

form quad and trying to solve. M1

$$x-3 = \pm 4 \Rightarrow x = -1 \text{ or } 7$$

A1

$$\frac{d^2y}{dx^2} = \frac{1}{(x-3)^3}$$

$$\text{when } x = -1, \frac{d^2y}{dx^2} = \frac{1}{-4^3} < 0$$

nature investigation

M1

$$y = -\frac{1}{32} + \frac{1}{2} \times \left(-\frac{1}{4}\right) = -\frac{1}{32} - \frac{4}{32} = -\frac{5}{32}$$

subst. to get y.

$$\therefore \text{pt } \left(-1; -\frac{5}{32}\right) \text{ is maximum}$$

correct conclusion of max

A1

$$\text{when } x = 7, \frac{d^2y}{dx^2} = \frac{1}{(4)^3} > 0$$

$$\text{when } y = \frac{7}{32} + \frac{1}{2} \times \frac{1}{4} = \frac{7}{32} + \frac{4}{32} = \frac{11}{32}$$

A1 both correct.

$$\therefore \text{pt } \left(7, \frac{11}{32}\right) \text{ is minimum}$$

correct conclusion of min

A1

(c) On x-axis $\frac{1}{32}x + \frac{1}{2(x-3)} = 0$

equating to zero and attempting to form a quad

M1

$$x^2 - 3x + 16 = 0 \Rightarrow b^2 - 4ac = 9 - 64 = -55 < 0$$

Used the discriminant or quad graph

M1

Equation has no real root

$\therefore$ graph does not cross x-axis

A1

[13]

Subst. to get y

M1